

PARAGLIDING INTELLIGENCE PLATFORM

Technical System Architecture Document v2.0

A developer-ready architecture for a paragliding-first meteo, flight decision, safety, and XC intelligence platform

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Prepared for: Paragliding weather application concept originally defined as a Windy/Ventusky-class platform dedicated to paragliding, extended with paid tiers, AI decision intelligence, launch digital twins, XC planning, safety scoring, and club/school/event operations.

Document intent: Replace the previous high-level SAD with concrete technical service definitions, data ownership, API contracts, event flows, processing pipelines, schemas, algorithms, and implementation priorities. This is not filler text; it is a buildable architecture foundation.

0. Document Control

Field	Value
Document name	Paragliding Intelligence Platform - Technical System Architecture Document v2.0
Scope	System architecture, service boundaries, weather pipelines, scoring/risk engines, AI/ML architecture, data ownership, APIs, deployment model
Out of scope for this volume	Full UI wireframes, full OpenAPI file, full DDL migrations, operational runbooks, financial projections; these are separate later volumes
Primary users	Engineering team, product owner, investor technical reviewer, mobile/web developers, data/ML engineers
Quality target	Developer-ready foundation; enough detail to split work into epics and start implementation

1. Executive Technical Summary

The target system is not a generic weather map. It is a paragliding decision-intelligence platform. Generic weather products show layers; this platform converts weather, terrain, site rules, airspace, live observations, pilot skill, and historical local behavior into actionable GO/NO-GO recommendations and flight windows.

The architecture is built around seven technical pillars:

- Weather data fusion: ingest global models, Nordic/regional models, radar/lightning, live stations, webcams, and pilot reports.
- Paragliding site graph: launches, landings, hazards, wind-sector rules, rotor zones, valley effects, access notes, and legal restrictions.
- Forecast cube: normalized four-dimensional weather data indexed by latitude, longitude, altitude/pressure level, and time.
- Decision engines: flyability, XC potential, risk, route, and pilot-specific recommendation services.
- Digital twins: site-specific learning models that compare forecast versus real launch conditions and flights.
- Safety-first AI: LLM-generated briefings are grounded in structured data and cannot invent weather.
- Paid operational layers: PRO/PRO+, clubs, schools, competitions, rescue, and API licensing.

1.1 Competitive Positioning

Competitor capability	Typical market product	Required advantage in this platform
Animated weather maps	Windy-style layers, particles, model and level selectors	Keep comparable visualization, but add launch-specific decision scores and pilot-level logic
Large layer catalog	Ventusky-style many layers, radar, satellite, webcams	Do not compete only on layer count; compete on paragliding interpretation and site microclimate correction
Flight computer/tracking	XCTrack-like navigation and telemetry	Integrate weather intelligence before/during/after flight, not only navigation
Point forecast	Generic coordinate forecast	Convert forecast into flyability, rotor/foehn/valley wind risk, and site-specific launch window
Alerts	Generic weather threshold alerts	Pilot/site-specific alerts: "Vaga NW launch becomes safe for EN-B from 12:20 to 15:40"

1.2 Product Promise

The product should answer:

Can I fly today?

Where should I drive within 2 hours?

Which launch direction is safest?

When is the launch window?

What is the expected cloudbase and XC potential?

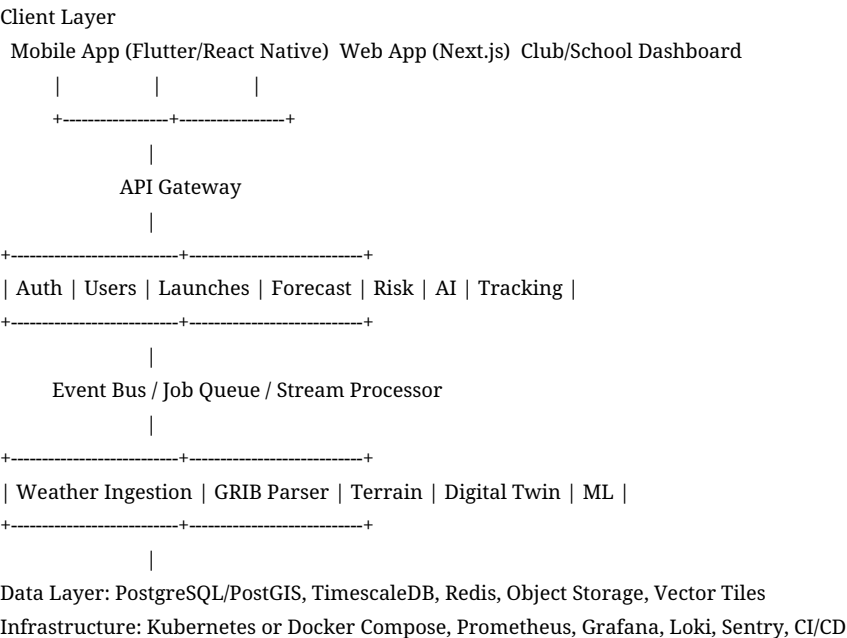
What can go wrong: gusts, rotor, overdevelopment, foehn, valley wind, airspace?

How confident is the recommendation and why?

2. Architecture Principles

Principle	Technical rule	Reason
Safety over engagement	When confidence is low, the system must degrade to CAUTION/NO-GO, not optimistic GO	Paragliding decisions can be safety-critical
Data-grounded AI	LLM can summarize only validated structured observations and forecast outputs	Prevents hallucinated weather advice
GIS-first	Every important object has geometry, elevation, CRS metadata, and spatial index	Paragliding is terrain-dependent
Forecast provenance	Every derived score stores source model, run time, processing version, and confidence	Auditability and model comparison
Offline-first mobile	Core maps, launch rules, airspace, and latest forecast cache must work offline	Pilots often have poor signal at launch
Composable services	Weather, risk, AI, tracking, billing, clubs/schools are independently deployable	Reduces coupling and permits staged development
Observe everything	Every ingestion job, score calculation, alert decision, and AI briefing has logs/metrics/traces	Needed for trust, debugging, and commercial operations
Graceful degradation	If radar fails, use model precipitation; if AI fails, show deterministic briefing; if live station fails, mark stale	Avoid broken user experience

3. High-Level System Architecture



3.1 Runtime Components

Component	Technology recommendation	Primary responsibility
Mobile app	Flutter or React Native	Map, offline forecasts, tracking, alerts, AI briefing, flight debrief
Web app	Next.js + MapLibre GL + deck.gl	Meteo map, dashboards, admin, club/school/event portals
API gateway	FastAPI gateway or Kong/Traefik	Routing, auth verification, rate limiting, API versioning
Weather services	Python/FastAPI + Celery/RQ/Arq	Ingestion, GRIB/NetCDF parsing, forecast normalization
GIS services	PostGIS + Tippecanoe/tegola/tileservr-gl/PMTiles	Launch geometry, airspace, DEM-derived layers, vector tiles
ML services	Python, scikit-learn/XGBoost/LightGBM, MLflow	Forecast correction, thermal prediction, risk classification
AI services	LLM gateway + RAG/context builder	Grounded natural-language flight briefings and coach
Tracking services	WebSocket/MQTT + TimescaleDB	Live telemetry, competition tracking, rescue mode
Object storage	S3/MinIO	GRIB, NetCDF, GeoTIFF, IGC, screenshots, forecast tiles
Cache	Redis	Hot forecasts, session cache, alert dedupe, tile metadata

3.2 Core Service Ownership

Service	Owns data	Consumes	Publishes
auth-service	users, identities, sessions, tokens	Billing entitlements	UserCreated, EntitlementChanged
launch-service	launches, landings, hazards, wind sectors, site metadata	Terrain, airspace	LaunchUpdated, SiteRuleChanged
weather-ingest-service	weather source runs, raw file metadata	External weather APIs	WeatherRunAvailable, RawFileStored
forecast-cube-service	normalized forecast cells, vertical profiles	Raw GRIB/JSON, terrain	ForecastCubeReady
terrain-service	DEM tiles, slope/aspect/ridge/valley derivatives	Copernicus/SRTM DEM	TerrainTileReady
risk-engine-service	risk scores, risk explanations	Forecast cube, launch rules, terrain	RiskScoreCalculated
flyability-service	flyability and XC scores	Risk, forecast, pilot profile	FlyabilityCalculated
digital-twin-service	site forecast errors, correction models, site learned parameters	Forecasts, stations, pilot reports, tracks	CorrectionModelUpdated
ai-briefing-service	briefing requests, grounded context snapshots	Forecast/flyability/risk/site/pilot	BriefingGenerated
tracking-service	live positions, flight tracks, IGC references	Mobile telemetry, airspace	TrackPointReceived, AirspaceWarning
notification-service	alert rules, delivery attempts	Scores, weather updates, tracking events	NotificationSent
billing-service	subscriptions, entitlements, invoices	Stripe or PSP webhooks	EntitlementChanged

4. Weather Data Strategy

The platform must support both rapid MVP data access and long-term raw model processing. The MVP should use Open-Meteo and MET Norway for fast implementation. The commercial version should add ECMWF/GFS raw processing and regional models where licensing allows.

4.1 Source Catalog

Source	Role	Resolution/notes	Integration mode	MVP?
Open-Meteo	Fast forecast API and historical/reanalysis access	Convenient JSON point forecast, multiple models and up to 16 forecast days depending endpoint/options	REST pull by coordinate/bounding grid	Yes
MET Norway Locationforecast	Nordic high-quality point forecast	Forecast for coordinates; most detailed for Nordic region; not suitable above-ground aviation profiles per docs	REST pull for site points	Yes for Norway/Nordics
ECMWF Open Data	Primary global model and AIFS/IFS medium range	Open real-time data with rolling archive; commercial use subject to terms/attribution	GRIB pull to object storage	V1/V2
NOAA GFS via NOMADS	Fallback/global model and ensemble baseline	GFS 0.25 degree data available through NOMADS GRIB filters	GRIB2 partial download	V1
Radar/Nowcast	Rain cells and short-term precipitation risk	Country-specific APIs; MET Norway Nowcast for Nordic short-term precip	REST/raster ingest	V1
Lightning	Thunderstorm real-time risk	Commercial feeds or public regional feeds where available	Stream/API	V1/V2
Live wind stations	Truth layer for launches and landings	IoT devices, club stations, public stations	MQTT/HTTP webhook	MVP+
Webcams	Visibility/cloud/windsock reality layer	Launch webcams, public webcams, own camera network	Image pull + CV analysis	V2

4.2 Weather Ingestion Pipeline

Schedule trigger or source webhook

- > weather-ingest-service creates weather_runs row
- > downloader stores raw file to s3://weather-raw/{source}/{model}/{run}/...
- > parser job validates checksum, variable coverage, forecast hours
- > forecast-cube-service normalizes variables and vertical levels
- > derived-weather-service computes wind profiles, cloudbase, thermal inputs
- > risk-engine-service recalculates launch windows for affected sites
- > notification-service evaluates user alert rules

4.3 Forecast Cube Data Model

The forecast cube is the canonical normalized model output. It can be implemented as chunked Zarr/NetCDF in object storage for large gridded data, with queryable metadata and hot point/area summaries in PostgreSQL/TimescaleDB.

Dimension	Description	Example
model_run	Source model and run time	ECMWF IFS 2026-06-04T00Z
valid_time	Forecast valid timestamp	2026-06-05T14:00Z
x/y	Geographic grid indexes or lon/lat	10.713E, 61.875N
vertical	Surface, height above ground, pressure level, or model level	10m AGL, 925 hPa, 850 hPa
variable	Normalized weather variable	u_wind_ms, v_wind_ms, gust_ms, cloud_cover_pct
value	Float value with units normalized	6.4
quality	Source status, interpolation status, confidence	native, interpolated, stale, missing

4.4 Variable Normalization

Canonical variable	Unit	Derived from	Used by
wind_speed_ms	m/s	$\sqrt{u^2+v^2}$	Map, flyability, risk, alerts
wind_dir_deg	degrees meteorological	atan2 conversion from u/v	Launch direction matching

gust_ms	m/s	native gust or corrected estimate	Safety, alerting
wind_shear_ms_per_100m	m/s/100m	vertical profile gradient	Turbulence/risk
cloudbase_m_agl	m AGL	LCL estimate or model cloud base	XC and safety
thermal_strength_ms	m/s	solar + lapse + terrain model	XC score
precip_mm_h	mm/h	native precip/radar	NO-GO/rain alert
cape_j_kg	J/kg	native instability variable	Overdevelopment/storm risk
visibility_m	m	model or station/webcam estimate	Safety
pressure_hpa	hPa	MSLP or surface pressure	Trend/föhn clues

4.5 Update Frequency

Pipeline	Frequency	SLA target	Storage TTL
Open-Meteo point refresh	Every 30-60 min for favorite/visible sites	90% complete under 3 min	16 days hot; 90 days aggregates
MET Norway point refresh	Hourly for Nordic active sites	90% complete under 3 min	9 days hot; 90 days aggregates
ECMWF/GFS GRIB run ingest	Every model cycle; initially 00/12Z, later 00/06/12/18Z	Raw stored within 45 min of availability	Raw 7-14 days, derived 90 days, skill metrics permanent
Radar/nowcast	5-15 min where available	Under 2 min delay after source update	48h hot
Weather station observations	1-5 min	Under 30 sec if own MQTT station	Permanent daily aggregates; raw 1 year
Pilot reports	Real-time	Under 10 sec	Permanent, moderation-aware

5. Terrain, GIS, and Airspace Architecture

5.1 Terrain Data Pipeline

Terrain is a first-class data source because ridge lift, rotor, thermal triggers, wind acceleration, and safe landing are terrain-dependent.

DEM source tiles (Copernicus GLO-30/GLO-90, SRTM fallback)

- > terrain-ingest-service validates CRS and tile coverage
- > raster processing computes: elevation, slope, aspect, ruggedness, ridge lines, valley lines
- > vectorization creates ridge/valley polygons and slope-aspect grids
- > PostGIS stores derivatives near launches; object storage stores GeoTIFF/COG tiles
- > tile-service publishes vector/raster overlays to client

Derivative	Computation	Use
Slope	Gradient from DEM	Thermal trigger probability, landing difficulty
Aspect	Slope facing direction	Solar heating, launch suitability
Ridge line	Topographic prominence/curvature extraction	Ridge lift and rotor candidates
Valley axis	Flow accumulation / terrain skeleton	Valley wind risk and route planning
Terrain exposure	Upwind openness relative to wind direction	Launch wind representativeness
Landing field slope	DEM + landcover + obstacle mask	XC safe landing suggestions

5.2 GIS Storage Rules

All geometries stored as SRID 4326 in public API.

Analysis jobs may use projected CRS per region for distance/area accuracy.

PostGIS indexes:

GIST geometry indexes on launch, landing, airspace, hazard, station, track segments.

BRIN indexes on time-series tables by timestamp.

Composite indexes for (site_id, valid_time), (model_run_id, valid_time), (user_id, created_at).

5.3 Airspace Guardian

Data object	Fields	Behavior
airspace_zone	geometry, lower_altitude, upper_altitude, class, activation schedule, source, effective date	Rendered on map and queried by tracking service
notam_event	geometry/area, active_from, active_to, text, source	Alerts only when confidence and source permit
airspace_warning	track_id, zone_id, distance_m, altitude_margin_m, severity, timestamp	Mobile push/voice warning and competition dashboard

Tracking point received

- > airspace-service finds zones within horizontal buffer
- > altitude resolver compares GPS/pressure altitude to zone limits
- > warning rules apply hysteresis to avoid spam
- > mobile receives voice-safe short warning: "CTR ahead, 900 meters, turn left or descend"

6. Launch Site Domain Model

The launch site database is the differentiator. Generic weather apps forecast coordinates. This platform models a site as an operational flying object with safe wind sectors, hazards, local rules, live evidence, and learned behavior.

6.1 Core Site Entities

Entity	Purpose	Key fields
site	Logical flying area	id, name, country, region, status, difficulty, owner/club, source, verification_status
launch	Actual takeoff point	site_id, geometry, elevation_m, orientation_deg, slope_deg, surface_type, min/max wind, gust limit
landing	Normal/emergency landing field	geometry, elevation_m, size_m2, slope_deg, obstacles, legal_status
wind_sector_rule	Launch-safe wind directions	launch_id, direction_from_deg, direction_to_deg, min_ms, max_ms, max_gust_ms, pilot_level
hazard	Powerlines, trees, rotor zones, restricted roads, livestock	geometry, severity, type, note, active flag
site_observation	Pilot or sensor reality input	site_id, source, wind, gust, cloudbase, visibility, thermal, report_quality
site_digital_twin	Learned correction profile	site_id, model_version, correction_features, confidence, last_trained_at

6.2 Launch Rule Example

```
{
  "launch_id": "vaga-nw-main",
  "rule_type": "wind_sector",
  "pilot_level": "intermediate",
  "direction_from_deg": 285,
  "direction_to_deg": 35,
  "min_wind_ms": 2.0,
  "max_wind_ms": 6.5,
  "max_gust_ms": 8.5,
  "max_gust_spread_ms": 3.0,
  "cloudbase_min_m_agl": 450,
  "rain_allowed": false,
  "notes": "Rotor risk with strong W over ridge shoulder; verify windsock and landing field before launch."
}
```

6.3 Site CRUD APIs

Endpoint	Method	Role	Response/behavior
/v1/sites	GET	Public/Pro	Nearby/filterable sites with current flyability summary
/v1/sites/{site_id}	GET	Public/Pro	Full site, launches, landings, hazards, rules, latest forecast
/v1/sites	POST	Moderator/Admin	Create proposed site, status=pending_verification
/v1/launches/{launch_id}/rules	PUT	Moderator/Admin	Replace launch rule set with audit entry
/v1/sites/{site_id}/observations	POST	Authenticated user/station	Submit pilot report or station reading
/v1/sites/{site_id}/score/timeline	GET	Pro	Hourly flyability/safety/XC timeline by pilot profile

7. Flyability, Safety, and XC Scoring Engines

7.1 Score Separation

Do not collapse all results into one number. Pilots need to know whether conditions are technically flyable, safe enough, and worth flying for XC.

Score	Meaning	Primary users	Can be high while others low?
Flyability Score	Whether the site can be flown from launch based on wind/rain/cloudbase/rules	All pilots	Yes - flyable but poor XC
Safety Score	Inverse risk: gust spread, rotor, storms, foehn, visibility, pilot mismatch	All pilots, schools, tandem	Yes - safe sled ride but no XC
XC Potential Score	Thermal strength, cloudbase, wind aloft, route continuity, landing fields	XC/competition pilots	Yes - good XC but unsafe launch conditions
Confidence Score	Model agreement, freshness, station match, digital twin skill	All users	Scores are suppressed when confidence is low

7.2 Deterministic Flyability Algorithm

Inputs:

launch_rule, pilot_profile, forecast_profile, radar, station_obs, digital_twin_correction

Hard NO-GO gates:

rain_intensity > threshold
thunderstorm/lightning risk > threshold
wind not in permitted sector for pilot level
wind_speed > launch max OR gust > launch gust limit
cloudbase below site minimum
visibility below site minimum
official closure or airspace restriction

Soft scoring:

wind_score = triangular score around ideal launch wind
direction_score = angular distance from launch orientation / safe sector center
gust_score = penalty for gust spread and gust/wind ratio
cloudbase_score = normalized to launch + ridge/route needs
thermal_score = positive for XC, capped for beginner safety
confidence_score = model agreement + station recency + digital twin skill

Output:

status in {NO_GO, CAUTION, GO, EXCELLENT}
flyability_score 0..100
safety_score 0..100
xc_score 0..100
explanation list sorted by impact

7.3 Score Weights by Pilot Type

Factor	Beginner EN-A	Intermediate EN-B	XC pilot	Tandem/School
Wind sector match	0.22	0.18	0.14	0.20
Wind speed ideality	0.22	0.18	0.12	0.20
Gust spread	0.22	0.18	0.12	0.22
Rotor/terrain risk	0.14	0.16	0.16	0.16
Storm/OD risk	0.10	0.14	0.18	0.12
Cloudbase/visibility	0.06	0.08	0.12	0.06
Thermals/XC lift	0.02	0.04	0.16	0.02
Model confidence	0.02	0.04	0.10	0.02

7.4 Risk Engine Rules

Risk	Inputs	Rule implementation	Output
Rotor	Wind direction/speed, ridge geometry, lee-side launch/route geometry, terrain roughness	Compute lee polygons downwind of ridge lines; scale by wind speed and stability; intersect launch/route	rotor_risk 0..100 + geometry
Foehn	Pressure gradient across mountain ranges, humidity drop, wind aloft, temperature differences	Regional rules per mountain chain + pressure delta thresholds; model agreement required	foehn_risk 0..100
Valley wind	Solar heating, valley axis, time of day, synoptic wind, station trend	Empirical valley-flow model trained per site; warn for late-day strengthening	valley_wind_risk 0..100
Overdevelopment	CAPE, CIN, lapse rate, humidity, radar/satellite trend, cloud growth	Nowcast plus model instability; increase risk when cloud growth detected	od_storm_risk 0..100
Wind shear/turbulence	Vertical wind profile, gusts, terrain acceleration, model PBL height	Shear thresholds and turbulence proxy	turbulence_risk 0..100

7.5 Explanation Object

```
{
  "site_id": "vaga-nw",
  "valid_time": "2026-06-05T13:00:00Z",
  "status": "GO",
  "flyability_score": 84,
  "safety_score": 78,
  "xc_score": 71,
  "confidence": 0.82,
  "top_positive_factors": [
    {"factor": "wind_direction", "message": "NW wind matches launch sector", "impact": +18},
    {"factor": "cloudbase", "message": "Cloudbase expected 650 m above launch", "impact": +12}
  ],
  "top_risks": [
    {"factor": "gust_spread", "message": "Gust spread increases after 16:00", "impact": -10},
    {"factor": "valley_wind", "message": "Valley wind may strengthen late afternoon", "impact": -7}
  ],
  "hard_gates": []
}
```

8. Digital Twin and Site-Specific Forecast Correction

The digital twin is the main long-term moat. It converts a generic forecast into a site-specific forecast by learning repeatable local errors: ridge acceleration, valley wind timing, launch exposure, station bias, cloudbase bias, and thermal under/over-forecasting.

8.1 Digital Twin Data Sources

Input	Granularity	Quality controls
Forecast snapshots	site x valid_time x model_run	Stored before actual outcome; immutable to avoid leakage
Live station observations	1-5 minute readings	Station calibration, stuck-sensor detection, recency
Pilot launch reports	Manual reports at launch/landing	Reputation score, consistency, duplicate detection
Flight tracks/IGC	per flight path	GPS sanity, altitude sanity, declared privacy level
Webcam CV features	5-15 min	Cloud obstruction confidence, visibility class, windsock direction if visible
Incident/near-miss reports	event-level	Moderated, anonymized, not used punitively

8.2 Correction Targets

Target	Model type	Feature examples	Evaluation metric
launch_wind_speed_error	Gradient boosting regression	raw wind, direction, stability, hour, slope exposure, model, station distance	MAE/RMSE by site and season
launch_wind_direction_error	Circular regression / sin-cos regression	u/v components, terrain aspect, pressure gradient	Mean angular error
gust_error	Quantile regression	gust forecast, thermal strength, PBL height, terrain roughness	Pinball loss p90 gust
cloudbase_error	Regression	temp/dewpoint spread, humidity, elevation, cloud cover	MAE meters
thermal_strength_error	Regression/classification	solar radiation, cloud cover, lapse rate, aspect	MAE + usable thermal classification
go_no_go_outcome	Classification	rules + corrected forecasts + reports	Precision on NO-GO/CAUTION

8.3 Leakage Prevention

Rule: Any forecast correction model must use only data available before the forecast valid time.

`feature_time <= prediction_generated_at <= user_decision_time < valid_time`

Implementation:

- store forecast_snapshot with model_run_time and ingestion_time
- store station_obs_time separately
- training builder uses asof joins with strict time cutoffs
- never use resolved flight outcome features to train live predictors unless they would have been known at decision time
- preserve feature snapshots for every recommendation shown to users

8.4 Digital Twin Service APIs

Endpoint	Method	Purpose
/v1/sites/{id}/twin/skill	GET	Show forecast error and correction skill for a site
/v1/sites/{id}/twin/correction	GET	Return current correction deltas for a forecast hour
/v1/internal/twin/train	POST	Trigger site or regional model training job
/v1/internal/twin/evaluate	POST	Backtest model version before promotion
/v1/sites/{id}/forecast/explain	GET	Explain raw forecast vs corrected forecast vs observations

9. Thermal and XC Planning Engine

9.1 Thermal Prediction

Thermal prediction combines numerical weather, terrain, solar geometry, surface properties, and local site history. For MVP, use a heuristic model. For V2, train a model using IGC climbs and pilot reports as labels.

```
thermal_potential = f(
    solar_radiation,
    cloud_cover,
    surface_temp - air_temp,
    lapse_rate_0_1500m,
    humidity,
    wind_speed_low_level,
    slope_aspect_alignment_to_sun,
    terrain_ruggedness,
    known_trigger_points,
    digital_twin_site_bias
)
```

Label examples from IGC:

climb_segment if vario_avg > 0.8 m/s for >= 60 sec and circular drift radius < threshold

thermal_strength_label = median climb rate in segment

9.2 XC Route Planning

Step	Algorithm	Output
Candidate launch windows	Flyability timeline + XC score	Best start times
Thermal field generation	Raster/grid of expected thermal sources and cloudbase	Thermal waypoints
Glide graph construction	Nodes=thermals/landings/waypoints, edges=glide feasibility	Route graph
Wind drift model	Wind by altitude and expected climb drift	Downwind/upwind correction
Landing field scoring	Landcover/slope/access/airspace constraints	Safe landing options
Route optimization	A*/Dijkstra/beam search with risk penalty	20km/50km/100km route candidates

9.3 XC Route API

POST /v1/xc/routes/plan

Request:

```
{
  "launch_id": "vaga-nw-main",
  "start_time": "2026-06-05T12:30:00Z",
  "pilot_profile_id": "...",
  "target_distance_km": 50,
  "route_type": "open_distance|triangle|out_and_return|competition_task",
  "risk_tolerance": "low|normal|high"
}
```

Response:

```
{
  "route_id": "...",
  "score": 78,
  "expected_distance_km": 52,
  "expected_duration_min": 145,
  "waypoints": [...],
  "thermal_segments": [...],
  "landing_options": [...],
  "risks": [...]
```


10. AI Decision Engine

The AI layer is a presentation and reasoning assistant, not the source of weather truth. It receives a structured context created by backend services and produces briefings, comparisons, and debriefs with citations to data snapshots.

10.1 Grounded Context Builder

User question: "Where can I fly tomorrow within 2 hours of Oslo?"

- > intent = site_discovery
- > geocoder resolves user/home area
- > launch-service returns sites within travel radius
- > forecast/flyability services compute timeline per site
- > airspace/risk services add restrictions and risk notes
- > entitlement service applies Pro/Free detail limits
- > context builder composes JSON only from backend data
- > LLM generates ranked answer with confidence and caveats
- > response validator checks no unsupported weather claims

10.2 AI Safety Guardrails

Guardrail	Implementation
No invented values	LLM prompt receives machine JSON; validator rejects numbers not in context
Conservative uncertainty	If confidence < threshold, final answer must include "uncertain/caution" recommendation
No guarantee language	Disallow "safe" as absolute; use "conditions appear within defined limits"
Pilot-level personalization	Context includes wing class, hours, training mode, tandem/school mode
Traceability	Store context_snapshot_id and response version for every briefing
Emergency redirection	If user asks in-flight danger question, provide land-as-soon-as-practical style guidance and emergency features

10.3 Briefing Output Schema

```
{
  "briefing_id": "...",
  "answer_type": "site_recommendation|site_briefing|xc_plan|debrief",
  "summary": "Best window appears to be 12:30-15:30 at Vaga NW for intermediate pilots.",
  "decision": "GO|CAUTION|NO_GO|INSUFFICIENT_DATA",
  "confidence": 0.81,
  "sites_ranked": [...],
  "hard_warnings": [...],
  "assumptions": ["Pilot profile: intermediate EN-B", "Travel limit: 2 hours"],
  "data_sources": ["ECMWF 00Z", "MET Norway 09:00 refresh", "station VAGA-W1"],
  "unsupported_claims_detected": false
}
```

10.4 AI Features by Tier

Feature	Free	Pro	Pro+	School/Club/Event
Basic site summary	Limited	Full	Full	Full
AI flight briefing	3/day	Unlimited personal	Unlimited + model comparison	Team/event briefings
AI site discovery	No	Yes	Yes + route/weather optimization	Yes + member/student filtering
AI XC coach/debrief	No	Limited	Full IGC debrief	Instructor/student review
AI competition briefing	No	No	Limited	Full task briefing

11. Live Tracking, Alerts, and Accident Prevention

11.1 Tracking Data Path

- Mobile GPS sample every 1-5 seconds when flying
- > local smoothing/filtering and battery-aware batching
 - > WebSocket/MQTT upload when connected
 - > tracking-service writes hot position to Redis and durable stream
 - > TimescaleDB stores track points/segments
 - > airspace guardian checks upcoming conflicts
 - > alert engine checks weather deterioration near track
 - > club/event/rescue dashboards subscribe to live state

11.2 Weather Deterioration In-Flight Alerts

Alert	Trigger	Delivery
Gust increase	Station/nowcast/model trend exceeds pilot/site threshold	Push + watch + voice
Thunderstorm growth	Lightning/radar/CAPE/OD risk rises within route buffer	Urgent push + voice: land soon
Cloudbase collapse	Cloudbase falls near route/landing options	Push + route advisory
Airspace conflict	Projected path intersects zone within time/distance threshold	Voice + map highlight
Lost pilot / overdue	No telemetry after planned landing window or manual SOS	Trusted contacts/club/rescue workflow

11.3 Alert Deduplication

- alert_key = hash(user_id, alert_type, site_or_track_id, severity_band)
- Redis SETNX alert_key TTL 30-120 min
- If new severity > old severity, escalate immediately.
- If conditions clear for hysteresis window, close alert and permit future alerts.

12. Community, Reports, Webcams, and Reality Layer

12.1 Pilot Reports

Report field	Validation
wind_speed/gust/direction	Range checks; compare to station/model; ask for confidence
cloudbase	Must be plausible vs launch elevation and model; optional photo evidence
thermal strength	Bucketed values for easier input; inferred from IGC when available
status: flying/grounded	Trust weighted by reputation and proximity
free text	Moderation + structured extraction for hazards

12.2 Reputation Scoring

```
reporter_score = f(  
  account_age,  
  verified_pilot/school/club_status,  
  historic_agreement_with_stations/other_reports,  
  report_completeness,  
  moderation_flags,  
  proximity_to_site_at_report_time  
)
```

Reality layer combines reports by weighted median, not simple average.

12.3 Webcam AI

CV task	Output	Use
Visibility classification	clear / hazy / fog / clouded-in	Launch briefing and safety score
Cloud growth trend	stable / building / overdeveloping	OD risk
Windssock detection	direction + movement confidence	Reality wind check
Precipitation detection	rain/snow/fog confidence	NO-GO validation
Launch occupancy	people/gliders present optional privacy-aware	Operational status, not individual identification

13. Paid Tier, Entitlements, and Commercial Architecture

13.1 Entitlement Model

Plan	Entitlements
Free	3-day forecast, limited layers, 5 favorites, basic alerts, community reports
Pro	10-16 day forecast where available, flyability timeline, unlimited favorites, advanced alerts, AI briefing
Pro+	Model comparison, digital twin correction, XC planner, AI debrief, advanced risk layers
Club	Member dashboard, shared alerts, site admin, shared stations, club reports
School	Instructor dashboard, student skill filters, training weather windows, daily briefings
Event	Competition task weather center, live tracking, safety board, public map
Rescue	Drift/search area estimation, last-known-position tools, operational dashboard
API	Rate-limited forecast/flyability/site/risk endpoints for partners

13.2 Billing Architecture

payment provider webhook

- > billing-service validates signature
- > subscription row updated
- > entitlement snapshot generated
- > EntitlementChanged event
- > API gateway cache invalidated
- > user receives new feature access immediately

13.3 Entitlement Enforcement

Layer	Control
API gateway	Rate limits and route-level permission checks
Service layer	Feature flags and plan limits in business logic
Client UI	Hide/lock premium features, but do not rely on client-only enforcement
Data layer	Tenant/club/school isolation via <code>organization_id</code> + RBAC
AI layer	Context detail limits and quota controls by plan

14. Database Design - Implementation Core

14.1 Core Tables

-- PostgreSQL/PostGIS extensions

CREATE EXTENSION IF NOT EXISTS postgis;

CREATE EXTENSION IF NOT EXISTS timescaledb;

CREATE EXTENSION IF NOT EXISTS pgcrypto;

```
CREATE TABLE users (  
  id uuid PRIMARY KEY DEFAULT gen_random_uuid(),  
  email text UNIQUE NOT NULL,  
  display_name text,  
  created_at timestamptz NOT NULL DEFAULT now(),  
  deleted_at timestamptz  
);
```

```
CREATE TABLE pilot_profiles (  
  id uuid PRIMARY KEY DEFAULT gen_random_uuid(),  
  user_id uuid REFERENCES users(id),  
  skill_level text CHECK (skill_level IN ('beginner','intermediate','advanced','tandem','instructor','competition')),  
  wing_class text,  
  hours_flown numeric,  
  risk_tolerance text DEFAULT 'normal',  
  home_location geography(Point,4326),  
  created_at timestamptz DEFAULT now()  
);
```

```
CREATE TABLE sites (  
  id uuid PRIMARY KEY DEFAULT gen_random_uuid(),  
  name text NOT NULL,  
  country_code char(2),  
  region text,  
  status text DEFAULT 'active',  
  difficulty text,  
  centroid geography(Point,4326),  
  verification_status text DEFAULT 'unverified',  
  created_at timestamptz DEFAULT now(),  
  updated_at timestamptz DEFAULT now()  
);
```

```
CREATE TABLE launches (  
  id uuid PRIMARY KEY DEFAULT gen_random_uuid(),  
  site_id uuid REFERENCES sites(id),  
  name text NOT NULL,  
  location geography(Point,4326) NOT NULL,  
  elevation_m numeric,  
  orientation_deg numeric,  
  slope_deg numeric,  
  surface_type text,  
  notes text  
);
```

```
CREATE TABLE wind_sector_rules (  
  id uuid PRIMARY KEY DEFAULT gen_random_uuid(),  
  launch_id uuid REFERENCES launches(id),
```

```

pilot_level text,
direction_from_deg numeric,
direction_to_deg numeric,
min_wind_ms numeric,
max_wind_ms numeric,
max_gust_ms numeric,
max_gust_spread_ms numeric,
cloudbase_min_m_agl numeric,
rain_allowed boolean DEFAULT false,
rule_priority int DEFAULT 100
);

```

14.2 Forecast and Score Tables

```

CREATE TABLE weather_runs (
  id uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  source text NOT NULL,
  model text NOT NULL,
  run_time timestamptz NOT NULL,
  ingested_at timestamptz DEFAULT now(),
  status text NOT NULL,
  raw_uri text,
  variables jsonb,
  UNIQUE(source, model, run_time)
);

```

```

CREATE TABLE site_forecast_hourly (
  site_id uuid REFERENCES sites(id),
  launch_id uuid REFERENCES launches(id),
  weather_run_id uuid REFERENCES weather_runs(id),
  valid_time timestamptz NOT NULL,
  wind_speed_ms numeric,
  wind_dir_deg numeric,
  gust_ms numeric,
  cloudbase_m_agl numeric,
  thermal_strength_ms numeric,
  precip_mm_h numeric,
  cape_j_kg numeric,
  visibility_m numeric,
  confidence numeric,
  features jsonb,
  PRIMARY KEY (launch_id, weather_run_id, valid_time)
);

```

```

CREATE TABLE flyability_scores (
  id uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  launch_id uuid REFERENCES launches(id),
  pilot_profile_id uuid REFERENCES pilot_profiles(id),
  weather_run_id uuid REFERENCES weather_runs(id),
  valid_time timestamptz NOT NULL,
  status text NOT NULL,
  flyability_score int CHECK (flyability_score BETWEEN 0 AND 100),
  safety_score int CHECK (safety_score BETWEEN 0 AND 100),
  xc_score int CHECK (xc_score BETWEEN 0 AND 100),
  confidence numeric,
  explanation jsonb,

```

```
    created_at timestamptz DEFAULT now()
);
```

```
SELECT create_hypertable('site_forecast_hourly','valid_time', if_not_exists => TRUE);
```

14.3 Index Strategy

```
CREATE INDEX idx_sites_centroid_gix ON sites USING GIST (centroid);
CREATE INDEX idx_launches_location_gix ON launches USING GIST (location);
CREATE INDEX idx_scores_launch_time ON flyability_scores (launch_id, valid_time DESC);
CREATE INDEX idx_weather_runs_model_time ON weather_runs (source, model, run_time DESC);
CREATE INDEX idx_forecast_launch_time ON site_forecast_hourly (launch_id, valid_time DESC);
CREATE INDEX idx_forecast_features_gin ON site_forecast_hourly USING GIN (features);
```

15. API Specification - Implementation Core

15.1 Public User APIs

Endpoint	Method	Auth	Purpose
/v1/me	GET	JWT	Current user, profile, entitlements
/v1/sites/nearby? lat=&lon=&radius_km=	GET	Optional	Nearby launch sites with summary
/v1/sites/{site_id}	GET	Optional	Site details, launches, landings, hazards
/v1/launches/{launch_id}/forecast	GET	Pro tier gating	Hourly forecast variables
/v1/launches/{launch_id}/flyability	GET	Pro tier gating	Flyability/safety/XC timeline by pilot profile
/v1/alerts/rules	POST	JWT	Create site/weather alert rule
/v1/reports	POST	JWT	Submit pilot report
/v1/ai/briefing	POST	JWT + quota	Generate grounded flight briefing
/v1/tracks	POST	JWT	Start flight track
/v1/tracks/{id}/points	POST	JWT	Upload track batch

15.2 Internal APIs and Events

Interface	Producer	Consumer	Payload
WeatherRunAvailable	weather-ingest-service	forecast-cube-service	source, model, run_time, raw_uri, variables
ForecastCubeReady	forecast-cube-service	derived-weather/risk/flyability	model_run_id, bbox, valid_range
SiteRuleChanged	launch-service	risk/flyability/notification	site_id, launch_id, changed_rules
ObservationReceived	report/station-service	digital-twin/risk	site_id, observation, quality
FlyabilityCalculated	flyability-service	notification/AI/cache	launch_id, valid_time, scores
TrackPointReceived	tracking-service	airspace/weather-alert	track_id, point, altitude, speed
EntitlementChanged	billing-service	gateway/user/AI	user_id/org_id, plan, limits

15.3 Error Model

```
{
  "error": {
    "code": "FORECAST_NOT_AVAILABLE",
    "message": "No forecast is available for this launch and time range.",
    "retryable": true,
    "details": {"launch_id":"...", "requested_range":"..."},
    "trace_id": "..."
  }
}
```

Error code groups:

AUTH_*, ENTITLEMENT_*, VALIDATION_*, FORECAST_*, AIRSPACE_*, TRACKING_*, AI_*, RATE_LIMIT_*, INTERNAL_*

16. Deployment Architecture

16.1 MVP Deployment

Single region, low-cost deployment:

- Hetzner VPS or dedicated server
- Docker Compose
- Nginx/Traefik reverse proxy
- FastAPI services
- PostgreSQL + PostGIS + TimescaleDB
- Redis
- MinIO local/S3 compatible storage
- Celery/RQ workers
- Prometheus + Grafana + Loki

Pros: low cost, easy to operate by one developer.
Cons: limited HA; use backups and documented recovery.

16.2 Scale Deployment

Kubernetes cluster:

- ingress-nginx/Traefik
- api-gateway deployment
- weather-ingest workers with node pools
- forecast-processing workers with CPU-optimized nodes
- AI service with provider fallback / optional GPU node
- tracking service with WebSocket autoscaling
- managed PostgreSQL or HA Postgres operator
- object storage external S3 or MinIO cluster
- Prometheus/Grafana/Loki/Tempo/Sentry
- ArgoCD for GitOps deployments

16.3 Service Scaling Rules

Service	Scale metric	Initial rule
api-gateway	RPS, p95 latency	HPA at 60% CPU or p95 > 300ms
weather-ingest workers	queue length, job age	Scale workers when job age > 5 min
forecast-processing	CPU, queue length	Batch per model run; spot/ephemeral workers
tracking-service	active websocket count	Scale when > 5k active connections per pod
ai-briefing-service	requests/min, provider latency	Queue + quota; provider fallback
notification-service	pending deliveries	Scale by queue depth and provider rate limits

16.4 Backups and Recovery

Data	Backup strategy	RPO/RTO target
PostgreSQL	Daily full + WAL archiving + point-in-time recovery	RPO 15 min, RTO 2 hours
Object storage	Versioned buckets + lifecycle + cross-region copy for critical data	RPO 1 hour, RTO 4 hours
Redis	Ephemeral except alert dedupe; no critical state	RPO acceptable loss
Vector tiles/derived rasters	Regenerate from source; cache only	RTO 24 hours
ML models	MLflow registry + object storage artifacts	RPO 1 hour, RTO 2 hours

17. Security, Privacy, and Safety Controls

17.1 Authentication and Authorization

Area	Design
Auth	OAuth2/OIDC login plus email/password optional; short-lived JWT access tokens, refresh token rotation
RBAC	Roles: user, pro_user, club_admin, school_instructor, event_admin, moderator, support, system_admin
Org isolation	Every club/school/event resource has organization_id; service layer enforces membership
API keys	Separate API partner keys with scoped endpoints, rate limits, revocation
Audit logs	Admin edits, site rule changes, billing entitlement changes, moderation decisions, AI briefings

17.2 Privacy

Risk	Control
Flight track reveals home/habits	User can private/club/public share; default private; delayed public display option
Live tracking exposes location	Explicit per-flight opt-in; rescue contacts configured by user
School/student data	Org-scoped access, instructor role, retention rules
Webcam privacy	No face recognition; process scene/weather features only; blur/low-res where needed
Incident reports	Anonymization and moderation workflow

17.3 Safety Disclaimer Implementation

Safety disclaimers should not be a passive footer only. The app must encode safety in product behavior: conservative gating, confidence display, stale-data warnings, and explicit “verify at launch” prompts.

When data stale:

```
if station_age > max_station_age: mark_station_stale = true
if forecast_run_age > model_age_limit: confidence -= penalty
if confidence < 0.55: status cannot exceed CAUTION
UI must show: "Forecast confidence is low; verify conditions at launch."
```

18. Observability and Operations

18.1 Metrics

Metric	Owner	Purpose
weather_ingest_success_rate	weather-ingest	Source health and SLA
forecast_processing_job_age_seconds	forecast-cube	Backlog detection
score_generation_latency_ms	flyability	User-facing responsiveness
station_data_age_seconds	station-service	Reality layer freshness
ai_unsupported_claims_count	ai-briefing	Guardrail health
alert_delivery_success_rate	notification	Critical notification reliability
tracking_active_sessions	tracking	Capacity planning
model_skill_mae_by_site	digital-twin/ML	Value of correction engine

18.2 Logs and Traces

Every user-visible recommendation carries:

- trace_id
- forecast_run_id
- scoring_version
- risk_engine_version
- digital_twin_model_version
- context_snapshot_id if AI was used

This makes every answer reproducible and auditable.

18.3 Operational Dashboards

Dashboard	Widgets
Weather Operations	Source status, run freshness, job backlog, failed variables, map of stale regions
Safety/Alerts	Alerts generated, delivered, failed, escalated, stale stations, high-risk sites
AI Quality	Briefing volume, rejected unsupported claims, latency, token cost, user feedback
Commercial	Subscriptions, churn, conversion, API usage, entitlement errors
Tracking/Event	Active flights, websocket health, airspace warnings, rescue mode events

19. Implementation Roadmap

19.1 MVP - 12 to 16 Weeks

Epic	Deliverable	Acceptance criteria
Core app shell	Next.js web map + mobile proof of concept	Map loads, launch pins, time slider
Launch database	Admin import + site/launch/landing/rules model	At least 50 validated sites in one launch country/region
Weather MVP	Open-Meteo + MET Norway point forecast	Hourly forecast per launch; cache and provenance stored
Flyability v1	Deterministic scoring engine	Scores generated for all MVP launches; hard NO-GO gates work
Alerts v1	Favorite site alerts	User receives alert when favorite launch becomes GO/CAUTION/NO-GO
AI briefing v1	Grounded summary from structured context	LLM output contains no unsupported numbers; stores context snapshot
Paid tier foundation	Stripe/PSP entitlement model	Free/Pro feature gates enforced server-side

19.2 V1 Commercial

Epic	Deliverable
Model comparison	ECMWF/GFS/Open-Meteo/MET comparison in UI and scoring confidence
PostGIS terrain	DEM-derived slope/aspect/ridge data around sites
Risk v1	Rotor candidate zones, gust spread, OD/storm risk
Pilot profiles	Beginner/intermediate/XC/tandem scoring variations
Club accounts	Shared sites, reports, member dashboards
Mobile tracking	Live flight recording and airspace warning MVP

19.3 V2 Differentiator

Epic	Deliverable
Digital twin correction	Site-specific forecast correction using station/reports/tracks
Thermal map	Thermal triggers and XC potential map
Webcam reality layer	Visibility/cloud/windsock CV for selected sites
XC planner	Route graph, landing options, route scoring
School platform	Student skill filters, daily training briefing
Competition mode	Task weather, live tracking, safety dashboard

19.4 V3 Moat

Epic	Deliverable
Global site digital twins	Self-learning corrections for thousands of launches
Rescue tools	Drift/search estimation and emergency operations dashboard
API product	Partner access for flyability, risk, forecast, site metadata
Hardware network	Branded weather stations and station cloud subscription
AI coach	IGC debrief, mistakes, speed-to-fly, thermal decisions

20. Feature-by-Feature Technical Checklist

Feature	Technical implementation
Animated Weather Map	MapLibre GL + deck.gl/WebGL particles; vector/raster tiles cached; layer metadata API; time/altitude slider; tile cache invalidation per model run.
Launch Site Forecast	Point extraction from forecast cube; launch correction; hourly score timeline; provenance stored per score.
Flyability GO/NO-GO	Hard gates + weighted score; pilot profile variants; explanation object; confidence suppression.
Thermal Prediction	Heuristic v1, ML v2 using IGC climb labels; terrain/solar/cloud/lapse features.
XC Planner	Graph search over thermal nodes/landing fields; wind drift; airspace constraints; risk-aware route score.
Risk Engine	Rotor/foehn/valley wind/OD/turbulence models; outputs numeric risk + geometry + explanations.
Digital Twin	As-of feature store; forecast error labels; model registry; per-site skill display; no leakage.
AI Briefing	Intent parser + context builder + LLM + response validator + audit snapshot.
Pilot Reports	Weighted reality layer; reputation score; moderation; structured fields.
Webcam CV	Weather-only computer vision; no face recognition; visibility/cloud/windsock features.
Airspace Guardian	OpenAIP/national sources; PostGIS query; altitude logic; voice warning.
Live Tracking	Mobile telemetry; Redis hot state; Timescale durable track; WebSocket dashboards.
Alerts	Rule engine; dedupe; severity escalation; push/email/SMS/provider abstraction.
Club Platform	Organization RBAC; shared sites, stations, reports, safety board.
School Platform	Student profiles, instructor workflow, training window safety filter.
Competition Mode	Task planning, live tracking, weather board, safety alerts, scoring integration.
Billing	Plan entitlements, server-side enforcement, provider webhooks, API quotas.
API Product	Scoped API keys, partner rate limits, forecast/flyability/risk/site endpoints.

21. Source Notes and External Reference Points

This architecture is designed to exceed generic weather-map products by adding site-specific paragliding decision intelligence. The following external references informed the platform assumptions and integration choices:

- Windy API documentation describes map forecast visualizations with layers, particles, isolines, model/level controls, and point forecast APIs.
- Ventusky product pages describe broad consumer weather capabilities including many weather maps, radar, lightning, webcams, and multi-model forecasts.
- Open-Meteo documentation provides REST forecast APIs and historical/reanalysis access suitable for MVP and prototyping.
- ECMWF Open Data provides real-time forecast data access and a rolling archive; use with proper terms and attribution.
- MET Norway Weather API provides freely available weather data under Creative Commons terms; Locationforecast serves coordinate forecasts for the next nine days, with Nordic detail.
- NOAA/NCEP NOMADS provides access to datasets including GFS 0.25 degree GRIB products and GRIB filtering.
- OpenAIP provides API access to aeronautical data useful for airspace-related features.
- Copernicus DEM provides digital surface/elevation models in 90m/30m/10m variants depending geography, useful for terrain-derived paragliding features.

22. Immediate Engineering Backlog

Priority	Task	Owner role	Definition of done
P0	Create monorepo with backend/frontend/mobile/service folders	Tech lead	CI runs lint/test; Docker Compose boots API + DB + Redis
P0	Implement site/launch/landing/wind rule schema	Backend/GIS	CRUD endpoints + PostGIS indexes + seed data
P0	Implement Open-Meteo/MET Norway forecast import	Weather engineer	Forecast cached for site/time with source provenance
P0	Implement deterministic flyability v1	Backend/data	Score timeline endpoint returns status, scores, explanations
P0	Implement map UI with launch pins and timeline	Frontend	Map displays sites, forecast panel, score timeline
P1	Add user profiles and paid entitlement checks	Backend/frontend	Free/Pro gates enforced server-side
P1	Add alert rule engine	Backend/mobile	Favorite site notification delivered on score change
P1	Add AI briefing with grounding validator	AI/backend	No unsupported weather numbers in generated answer
P2	Add terrain ingestion and DEM derivatives	GIS/data	Slope/aspect/ridge data visible for pilot sites
P2	Add station/report reality layer	Backend/mobile	Reports affect confidence/reality display with reputation weighting

End of Volume 02 v2.0. Next documents should expand this into Volume 03 Database Design with full DDL, Volume 04 OpenAPI Specification, Volume 05 Weather Engine Design, Volume 06 AI/ML Design, and Volume 07 Mobile UX Specification.